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WEEK-END PRE-PROJECT PRESENTATION DATA SHEET

Domain-wise project calibration / evaluation framework. This helps evaluate whether a student **actually understands the project they built.**

For **Data Analytics, MERN Full Stack, and AI-ML projects.**

STUDENT NAME: Aditya Patil

DOMAIN NAME: Data Analytics

PROJECT NAME: Energy Consumption Pattern Analysis

1. Data Analytics Project Calibration Points

S. No.	Calibration Point	What Evaluator Checks	Example	Student Remark
1	Problem Statement	What business or data problem is being solved?	Sales performance analysis	The project analyzes energy consumption patterns to identify usage trends, peak consumption periods, and opportunities for efficient energy utilization.
2	Data Source	Where the dataset came from	Kaggle, company database, API	Dataset was collected from Kaggle containing household and industrial energy consumption records with time-based usage data.
3	Data Size & Structure	Rows, columns, type of data	50k rows, 12 columns	121273 rows, 2 columns
4	Data Cleaning	Handling null values, duplicates, outliers	Missing values replaced	Removed duplicate records, handled missing energy readings, corrected timestamp formatting, and treated outliers in consumption data.
5	Data Transformation	Feature engineering, normalization	Created revenue column	Created new features such as hourly usage averages, peak consumption categories, and daily/monthly energy summaries for analysis.

CIN: U62099UP2025PTC2177

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S. No.	Calibration Point	What Evaluator Checks	Example	Student Remark
6	Exploratory Data Analysis (EDA)	Patterns, correlations identified	Region vs Sales trend	Analyzed trends in energy usage across different time periods and identified correlations between environmental conditions and electricity consumption.
7	Tools Used	Why those tools were selected	Python, SQL, Power BI	Python was used for preprocessing and analysis, Pandas and NumPy for data manipulation, and Matplotlib/Seaborn for visualization of energy trends.
8	Visualization	Quality of dashboards/charts	Interactive Power BI dashboard	Created line graphs, heatmaps, bar charts, and time-series visualizations to represent consumption patterns and peak usage periods.
9	Insights Generated	Key insights from data	Q4 sales drop in north region	Peak energy consumption was observed during evening hours. Higher appliance usage significantly increased electricity demand during specific periods.
10	Business Impact	How insights help business	Inventory optimization	The analysis can help utility companies and consumers optimize electricity usage, reduce wastage, and improve energy efficiency planning.
11	Automation	Any automation done	Scheduled data refresh	Automated data preprocessing, trend analysis, and visualization generation using Python scripts in Google Colab.
12	Limitations	Dataset limitations	Incomplete historical data	The dataset did not include real-time smart meter data and had limited geographical diversity in energy usage records.
13	Future Scope	Improvements possible	Real-time analytics	Future improvements include real-time energy monitoring, AI-based energy demand forecasting, and integration with IoT-based smart grid systems.

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Universal Questions to Check if the Candidate Really Built the Project

Question	Purpose	Student Remark
Why did you choose this problem?	Checks understanding of problem	I chose this project because energy efficiency is becoming increasingly important, and data analytics can help identify patterns that reduce unnecessary energy consumption.
What challenges did you face?	Checks real development experience	The major challenges were handling time-series data, cleaning inconsistent timestamps, and identifying meaningful trends from large consumption datasets.
What would you improve if given more time?	Checks depth of thinking	I would develop predictive models for energy forecasting and create an interactive dashboard for real-time monitoring and analytics.
What alternative technologies could be used?	Checks technical maturity	Instead of Python and Colab, technologies like Power BI, Tableau, Apache Spark, or cloud-based analytics platforms could also be used.
How would you scale this to 1 million users?	Checks system design thinking	I would use distributed databases, cloud infrastructure, IoT integration, and scalable stream-processing systems to efficiently process large-scale energy data.

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